

HDF5 on the bench

A 6-step run from 25 °C to 50 °C on the real instruments, written in HDF5 by the new code, then checked file by file and compared against the same data in JSON.

28 Sep 2026, from 11:48 Folder C:\Users\spencer\Desktop\DATA\DLTS\092826\114655

2¹⁸ points × 100 reps Room temperature 30 °C

46 of 46 checks passed. Every file is complete, correctly typed and carries its attributes; Redo and Remove & Retake replaced only their own step; every analysis path gives identical results from HDF5 and from the same data in JSON; and Quick Analysis and Detailed Analysis give the same Arrhenius result on the run.

6.8×

smaller run folder (352 MB JSON → 52.1 MB HDF5)

32×

faster Extract & Average (5.18 s → 0.16 s)

0.15 s

mean HDF5 write per step during the run (JSON: 4.4 s)

10×

less memory reading one step (176 → 17 MB)

01 · THE RUN

What the instruments did

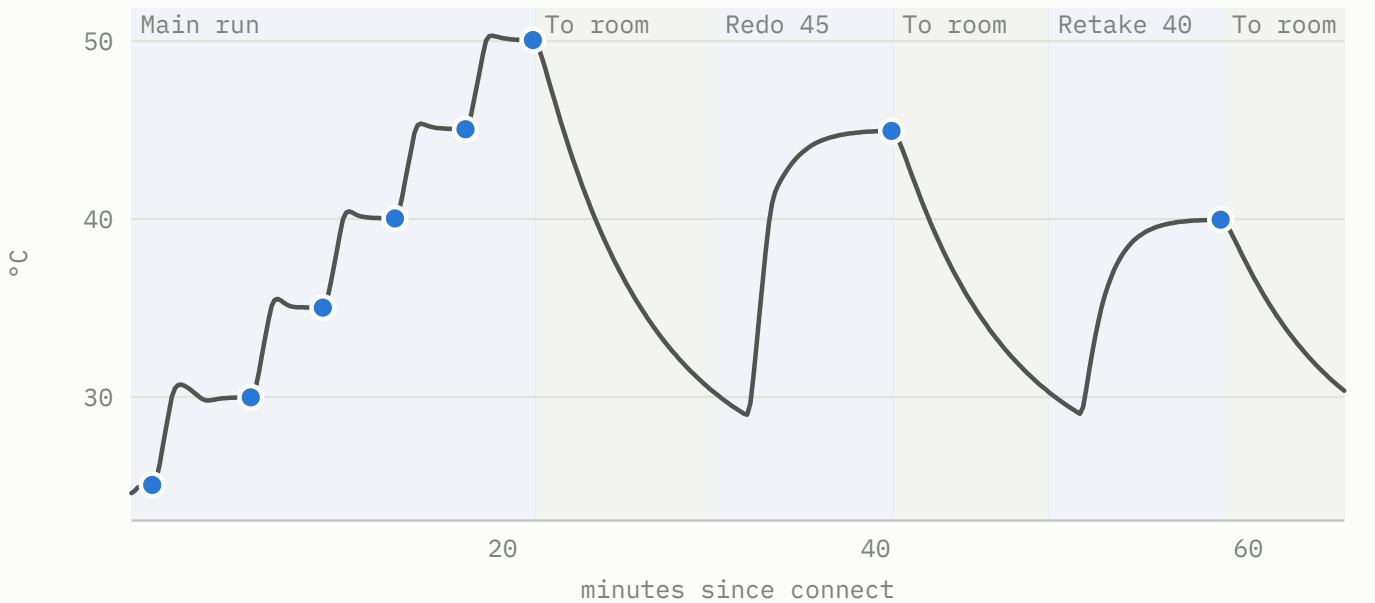
Driven headless through the GUI's own code path: init_experiment → run_experiment → finish_experiment, and the Room Temperature return after every run, then a Redo of 45 °C

and a Remove & Retake of 40 °C, each started from room temperature as the GUI's buttons allow.

Stage temperature during the session

Controller reading every 10 s. Shaded bands mark each phase; dots mark each acquisition, at the mean of the stage readings taken before and after it.

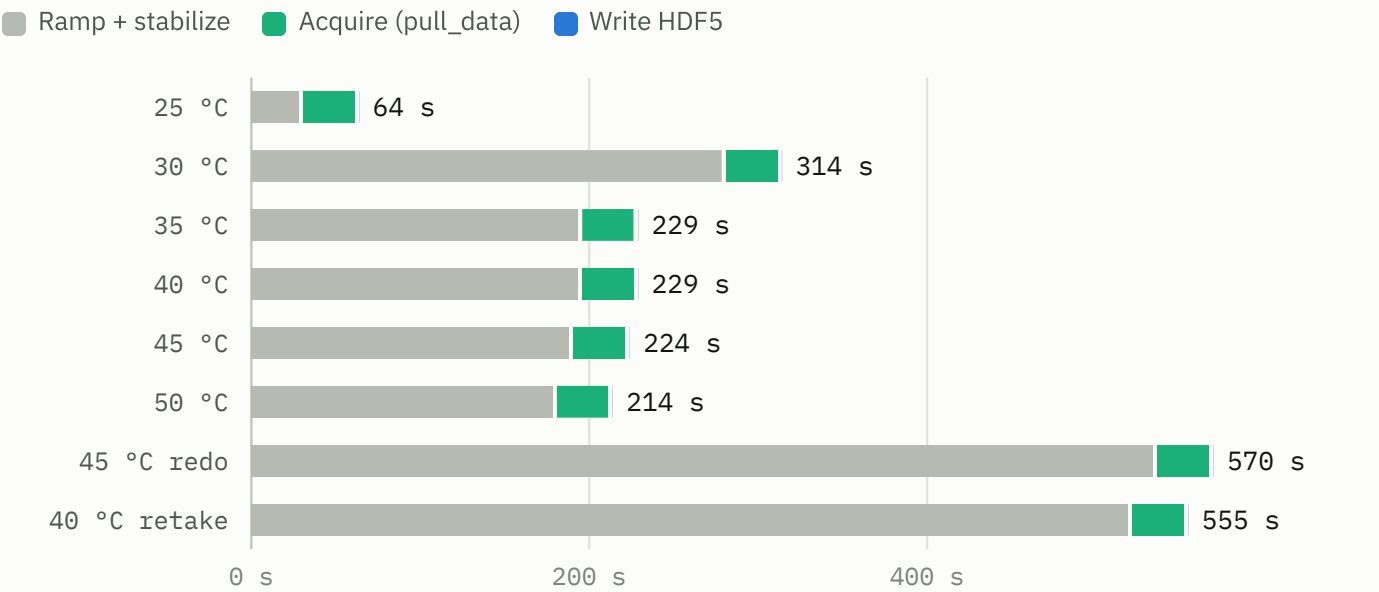
— Stage temperature ■ Acquisition



PHASE	START	END	DURATION
Main run	0.1 min	21.8 min	21.7 min
To room	21.8 min	31.4 min	9.7 min
Redo 45	31.4 min	40.9 min	9.5 min
To room	40.9 min	49.3 min	8.3 min
Retake 40	49.3 min	58.6 min	9.3 min
To room	58.6 min	65.1 min	6.5 min

Where each step's time went

Ramp and stabilization dominate. Writing the HDF5 file takes a fraction of a second per step.



STEP	STAGE T AT ACQUISITION	RAMP + STABILIZE	ACQUIRE	WRITE HDF5	FILE
25 °C	25.063 °C	31 s	33.1 s	0.153 s	p25p0.h5 · 8.66 MB
30 °C	29.984 °C	281 s	33.0 s	0.155 s	p30p0.h5 · 8.68 MB
35 °C	35.025 °C	196 s	33.0 s	0.157 s	p35p0.h5 · 8.68 MB
40 °C	40.037 °C	196 s	33.0 s	0.155 s	p40p0.h5 · 8.70 MB
45 °C	45.048 °C	191 s	33.0 s	0.154 s	p45p0.h5 · 8.68 MB
50 °C	50.059 °C	181 s	33.0 s	0.155 s	p50p0.h5 · 8.69 MB
45 °C redo	44.954 °C	537 s	32.9 s	0.157 s	p45p0.h5 · 8.68 MB
40 °C retake	39.966 °C	522 s	33.0 s	0.154 s	p40p0.h5 · 8.67 MB

02 · VERIFICATION

46 of 46 checks passed on the files the run wrote

FILE	8 CHANNELS, FLOAT64, GZIP + SHUFFLE	2^18 POINTS	TICK-STAMP AXIS INTACT	SETPOINT + STAGE T ATTRS	RUN_PARAMS ATTR	UNTOUCHED BY REDO/RETAKE
p25p0.h5	PASS	PASS	PASS	PASS 25.063 °C	PASS	PASS
p30p0.h5	PASS	PASS	PASS	PASS 29.984 °C	PASS	PASS
p35p0.h5	PASS	PASS	PASS	PASS 35.025 °C	PASS	PASS

FILE	8 CHANNELS, FLOAT64, GZIP + SHUFFLE	2^18 POINTS	TICK-STAMP AXIS INTACT	SETPOINT + STAGE T ATTRS	RUN_PARAMS ATTR	UNTOUCHED BY REDO/RETAKE
p40p0.h5	PASS	PASS	PASS	PASS 39.966 °C	PASS	n/a
p45p0.h5	PASS	PASS	PASS	PASS 44.954 °C	PASS	n/a
p50p0.h5	PASS	PASS	PASS	PASS 50.059 °C	PASS	PASS

PASS Run folder has one .h5 per temperature + runParams.txt, nothing else
 ['p25p0.h5', 'p30p0.h5', 'p35p0.h5', 'p40p0.h5', 'p45p0.h5', 'p50p0.h5',
 'runParams.txt']

PASS No leftover .h5.tmp files

PASS Redo at 45 C replaced the data (new acquisition time)
 2026-09-28T12:04:50 -> 2026-09-28T12:27:41

PASS Remove & Retake at 40 C replaced the data (new acquisition time)
 2026-09-28T12:01:03 -> 2026-09-28T12:45:20

PASS JSON copy of the real data round-trips losslessly

PASS Offline load (Live Tools): identical dataValues from JSON and HDF5

PASS Offline load (Live Tools): identical emissions + denoised output

PASS Offline load (Live Tools): an emission found at every temperature

PASS Qualitative Extract & Average: a transient for every temperature, no errors
 6 transients, reverse bias 6 ms; []

PASS Qualitative Extract & Average: identical transients from JSON and HDF5

PASS Detailed Analysis Load Data: every temperature loaded, identical from JSON and HDF5
 6 temperatures

PASS Quick Analysis and Detailed Analysis (Standard) give the same Et, sigma and Nt
 Et -1.036 eV in both, 4 fitted, 1 excluded

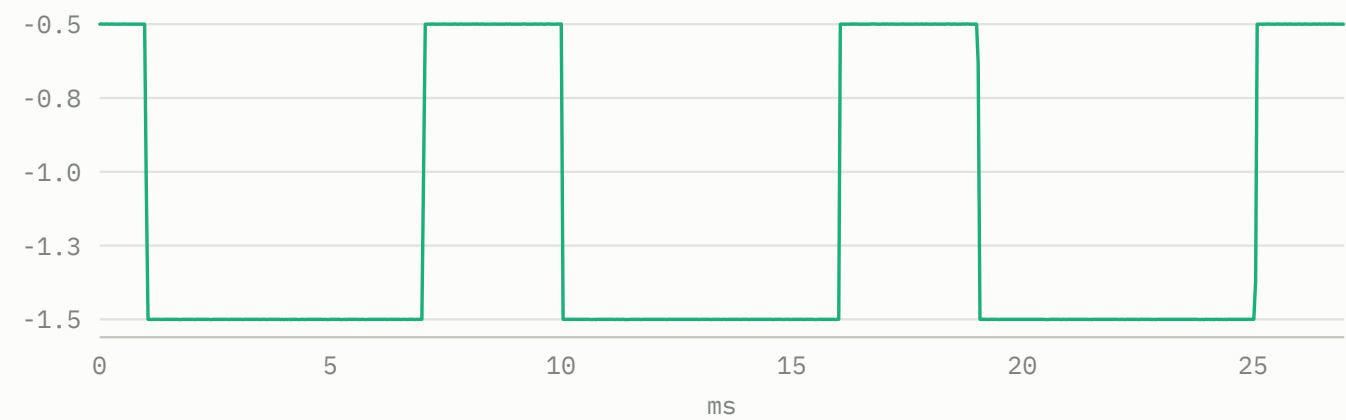
What was measured

MFIA settings: 501 kHz, 300 mV oscillation, 6 ms reverse bias and 3 ms fill (State Enable / Disable Time), Aux output between -0.5 V and -1.5 V. 2^18 points per step, averaged over 100 triggered repetitions.

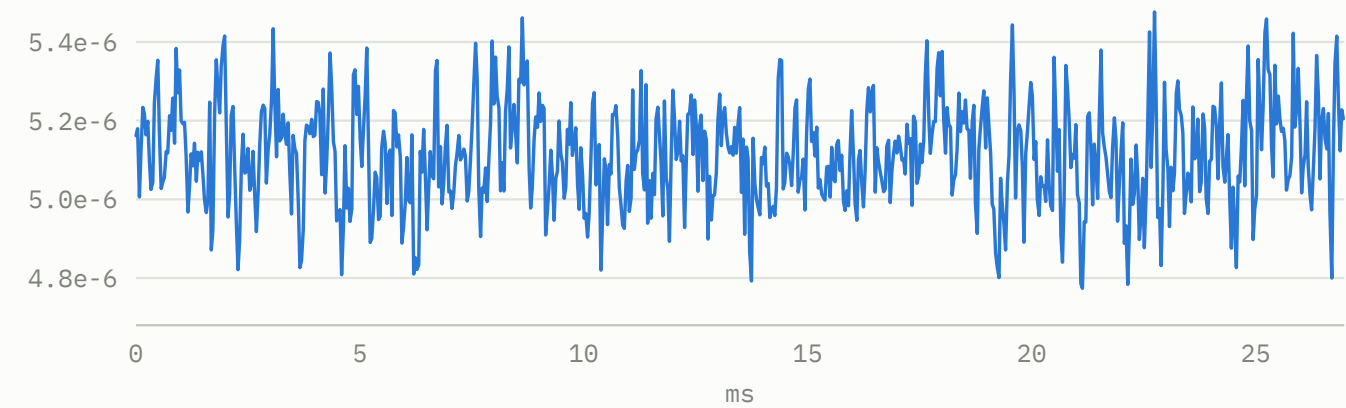
Raw acquisition, first three pulse cycles

25 °C

Excitation (Aux input 1, V)



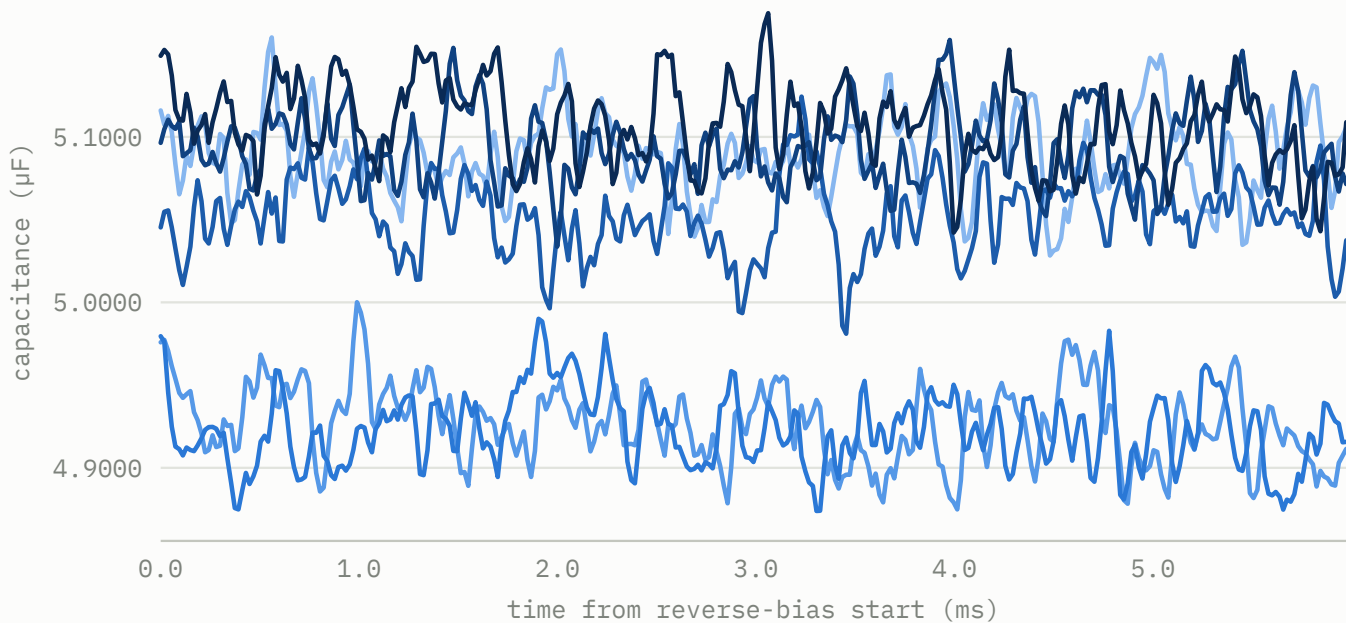
Impedance imaginary part as stored (ImpedanceIm)



Averaged transients from Qualitative "Extract & Average"

Every 6 ms reverse-bias cycle in each file, averaged, from the HDF5 files; the JSON copy gives identical curves. Pulses are located midway between the file's own excitation levels.

25 °C 30 °C 35 °C 40 °C 45 °C 50 °C

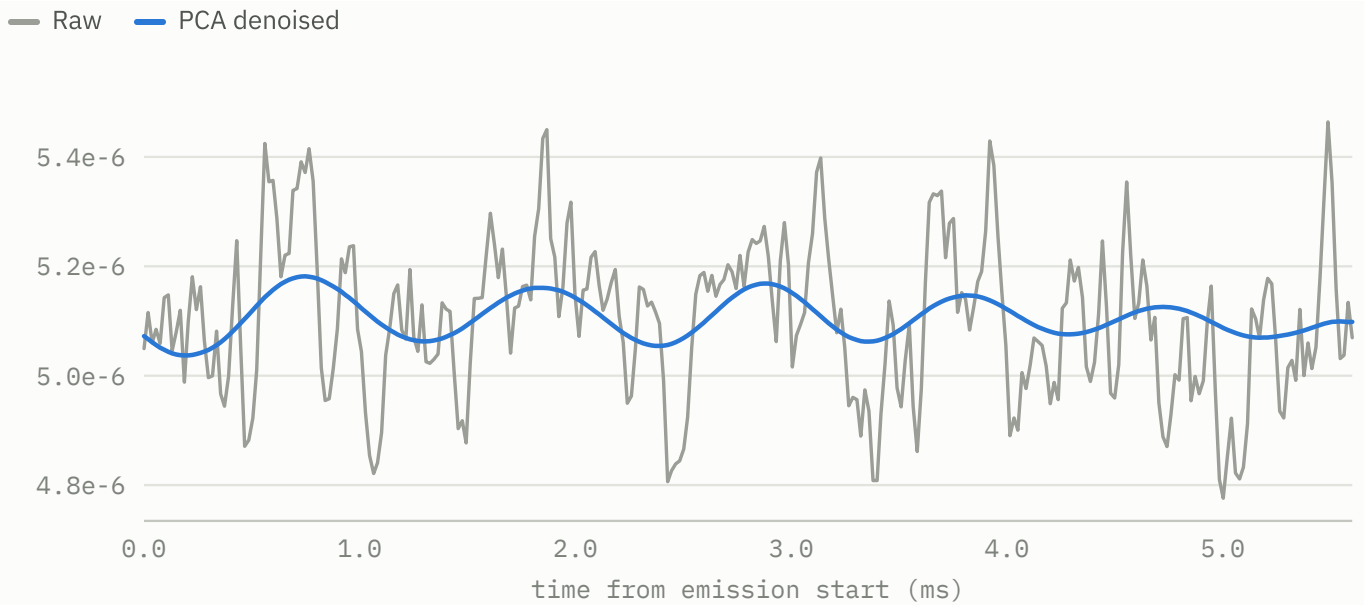


SETPOINT	POINTS PER CYCLE	C _∞ ESTIMATE (MF)	FIRST POINT (MF)	LAST POINT (MF)
25 °C	321	5.0984	5.1159	5.1057
30 °C	321	4.9611	4.9759	4.9114
35 °C	321	4.9201	4.9795	4.9157
40 °C	321	5.0563	5.0454	5.0375
45 °C	321	5.1288	5.0964	5.0714
50 °C	321	5.1289	5.1491	5.1089

Live Tools emission 0, raw vs PCA-denoised

Output of the offline pipeline (read_data → cleanup_data → selected_emissions → filter_emissions) on the HDF5 run.

25 °C



04 • REDO AND RETAKE

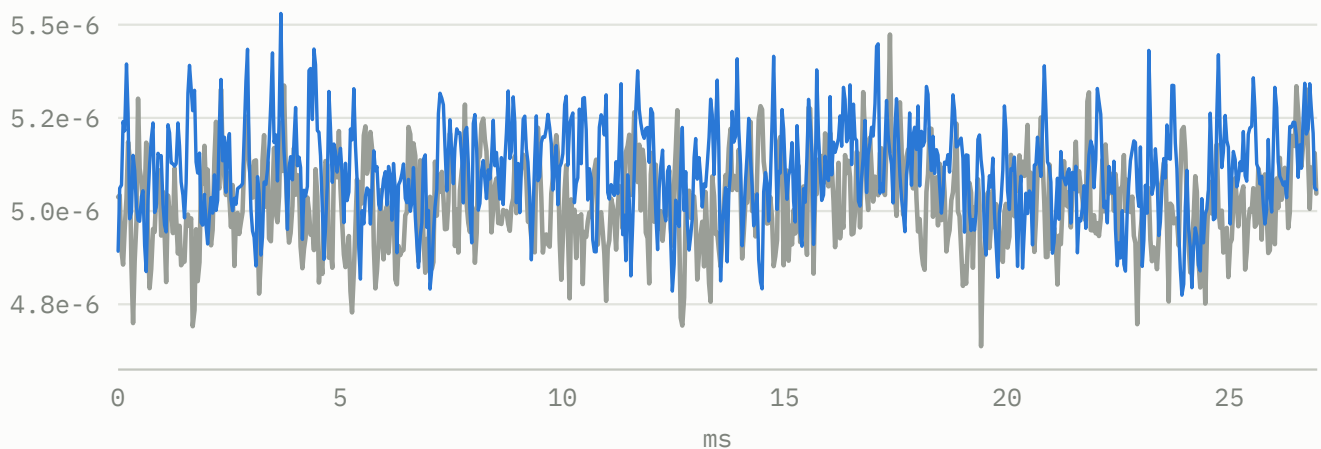
Re-acquired steps replaced only their own file

Each file was copied aside before being re-acquired, so the original and the replacement can be compared. The other four files were written once and left untouched.

Redo at 45 °C

Original 12:04:50 • replacement 12:27:41

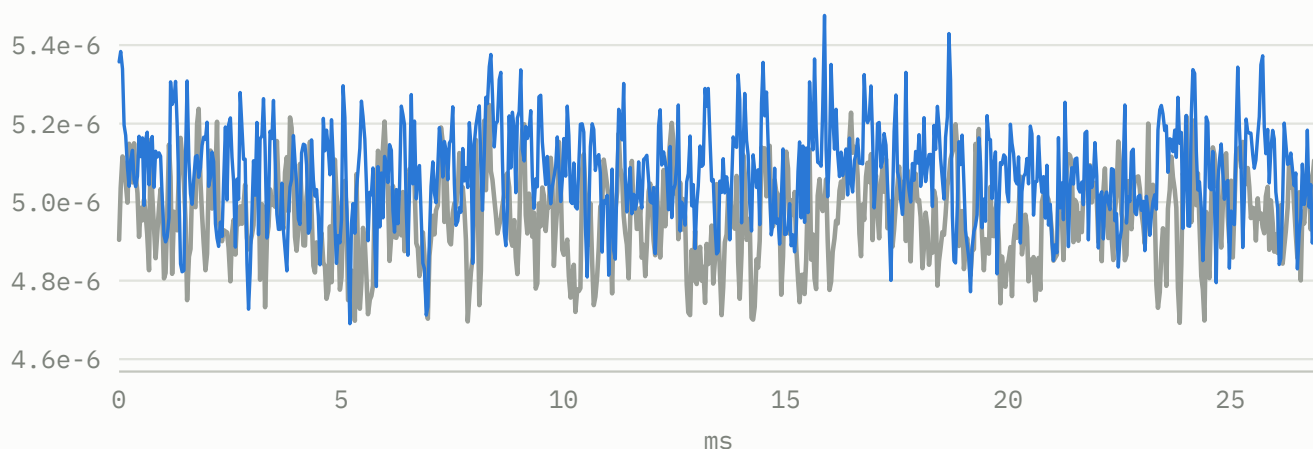
— Original acquisition — Replacement



Remove & Retake at 40 °C

Original 12:01:03 · replacement 12:45:20

— Original acquisition — Replacement



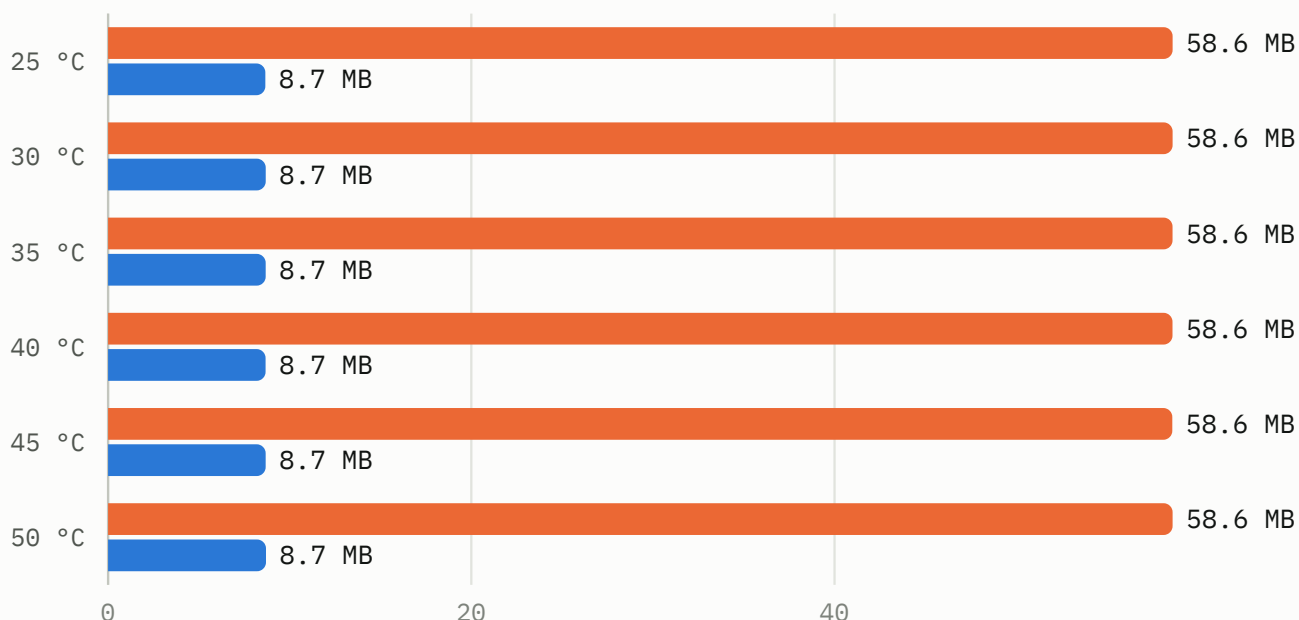
05 · JSON VS HDF5 ON REAL DATA

The evaluation's comparisons, repeated on this run

Each HDF5 file was also written as the JSON .txt the TXT format produces, from identical arrays. Both folders then went through the same analysis code.

File size per step

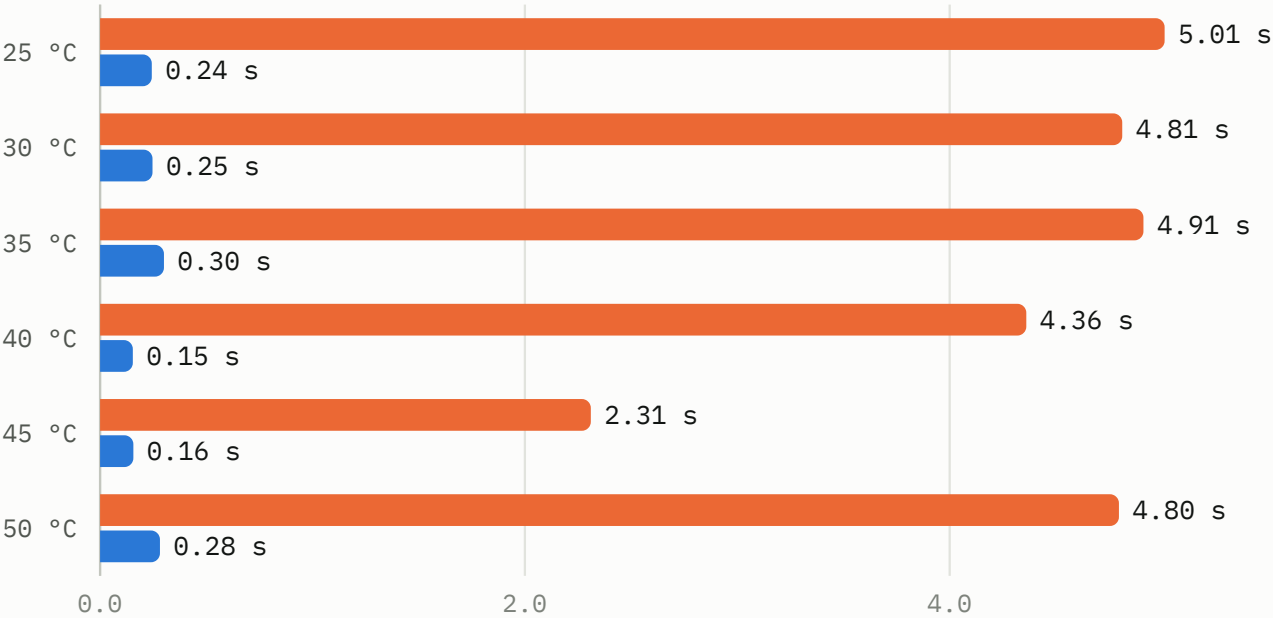
■ JSON .txt ■ HDF5 .h5



STEP	JSON (MB)	HDF5 (MB)	RATIO
25 °C	58.62	8.66	6.8×
30 °C	58.61	8.68	6.8×
35 °C	58.61	8.68	6.8×
40 °C	58.61	8.67	6.8×
45 °C	58.60	8.68	6.8×
50 °C	58.61	8.69	6.7×

Write time per step

JSON .txt
HDF5 .h5

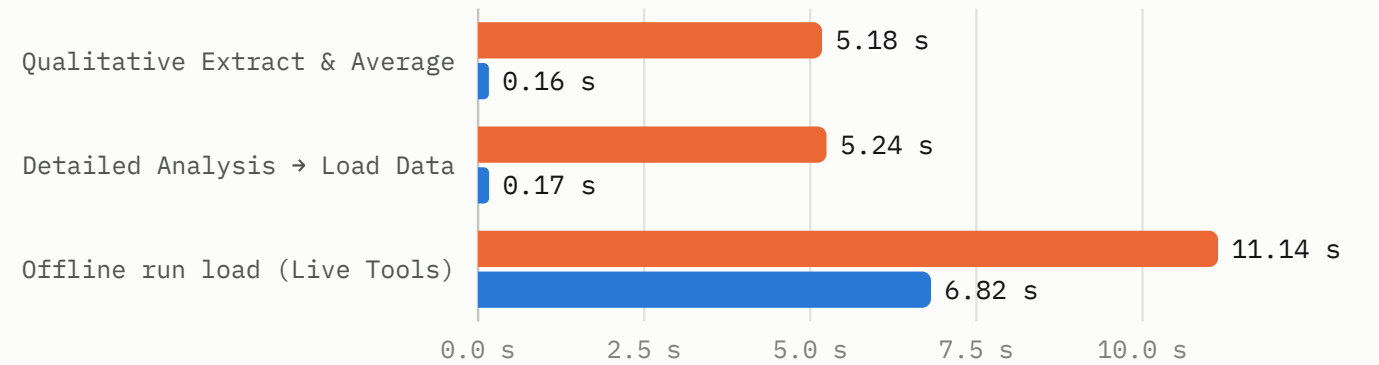


STEP	JSON WRITE (S)	HDF5 WRITE (S)	HDF5 DURING RUN (S)
25 °C	5.012	0.243	0.153
30 °C	4.812	0.247	0.155
35 °C	4.912	0.301	0.157
40 °C	4.361	0.154	0.155
45 °C	2.310	0.157	0.154
50 °C	4.797	0.282	0.155

Analysis paths, whole run

Wall-clock seconds for every temperature in the run.

JSON .txt HDF5 .h5

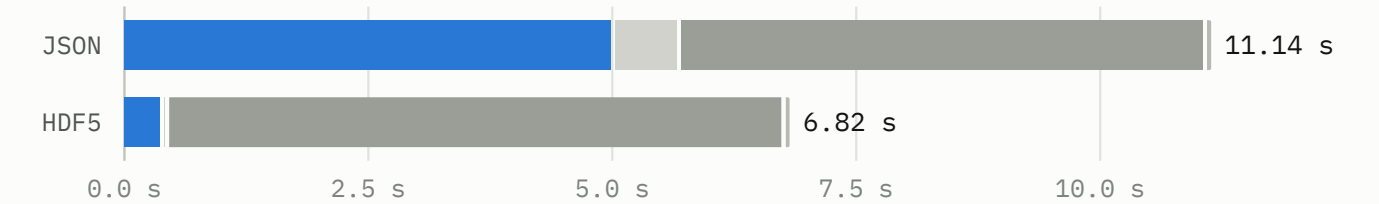


PATH	JSON (S)	HDF5 (S)	SPEED-UP
Qualitative Extract & Average	5.177	0.163	31.8×
Detailed Analysis → Load Data	5.244	0.166	31.6×
Offline run load (Live Tools)	11.137	6.817	1.6×

Offline run load by stage

Blue is file reading; greys are the stages HDF5 doesn't change.

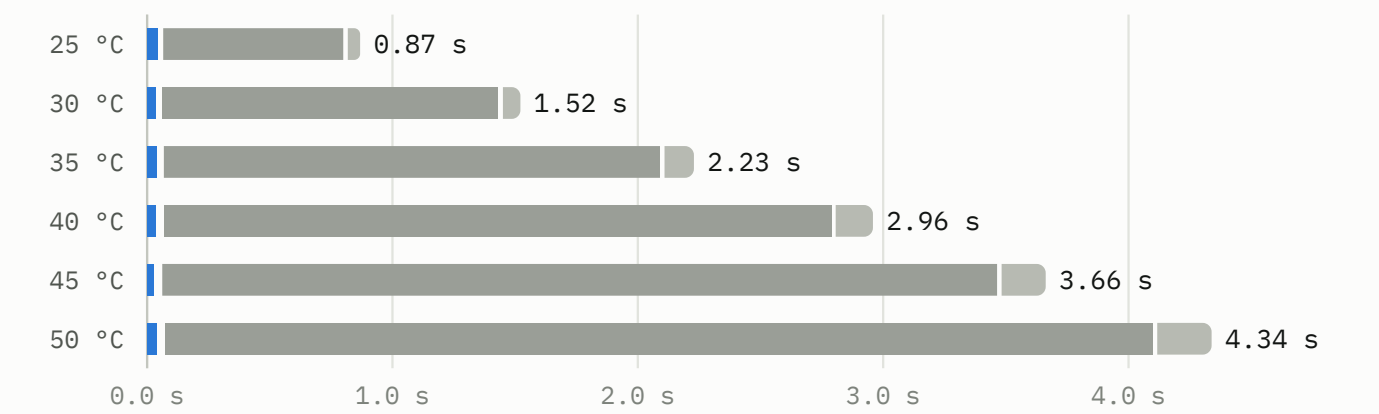
Read files Cleanup Clustering Denoise (PCA)



Live ingest of each new file during the run

The same work the Live Tools watcher does when a step's file appears.

Read file Cleanup Clustering Denoise + all emissions



FILE	TEMPS LOADED	READ (S)	CLEANUP (S)	CLUSTERING (S)	DENOISE + ALL (S)
p25p0.h5	1	0.060	0.006	0.75	0.05
p30p0.h5	2	0.053	0.010	1.39	0.07
p35p0.h5	3	0.057	0.012	2.04	0.12
p40p0.h5	4	0.054	0.014	2.74	0.15
p45p0.h5	5	0.045	0.016	3.42	0.18
p50p0.h5	6	0.056	0.018	4.04	0.22

Real run vs the evaluation's synthetic predictions

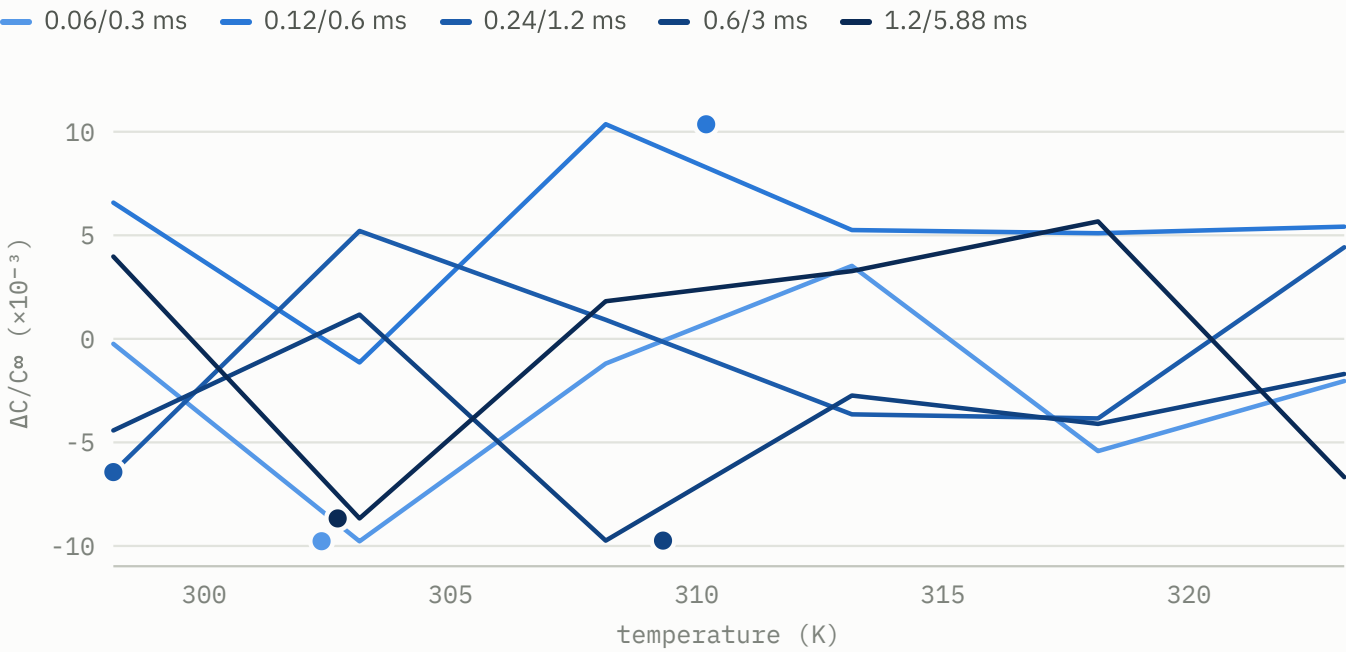
MEASURE	EVALUATION (SYNTHETIC, 40 × 2^16)	THIS RUN (REAL, 6 × 2^18)
Run folder size	560 → 66 MB (8.5×)	352 → 52.1 MB (6.8×)
Write per step	0.54 → 0.04 s (2^16 pts)	4.37 → 0.23 s (2^18 pts)
Qualitative Extract & Average	4.06 → 0.14 s (29×)	5.18 → 0.16 s (31.8×)
Detailed Analysis → Load Data	≈4.1 → 0.2 s (≈25×)	5.24 → 0.17 s (31.6×)
Offline run load	13.2 → 7.5 s (1.8×)	11.14 → 6.82 s (1.6×)
Offline: file reading only	4.3 → 0.6 s	5.03 → 0.41 s (12.3×)
Live ingest, last file	8.2 s; read 0.09 → 0.01 s	4.34 s; read 0.056 s (HDF5)
Peak memory, one step	42 → 4 MB (10×)	176 → 17 MB (10.5×)
Analysis results JSON vs HDF5	identical	identical

Quick Analysis and Detailed Analysis on this run

Windows 0.06/0.3, 0.12/0.6, 0.24/1.2, 0.6/3, 1.2/5.88 ms (the GUI's default windows scaled to this run's 6 ms reverse bias), peak search over the run's own range, 297.7 to 323.7 K. Both tabs give the same numbers, as they now share one pipeline. This run shows no trap emission: S(T) has no peak that moves to higher temperature for faster windows, and the fit gives $E_t = -1.04 \pm 2.26$ eV with $R^2 = -0.14$. 6 setpoints over 25 K with a 6 ms reverse bias were chosen to exercise the HDF5 path, not to characterize a trap. The result is shown only to confirm that the two tabs agree.

DLTS signal S(T) for each rate window

Lines are $\Delta C/C^\infty$ at each setpoint; a dot marks the peak each window reports. A peak on the edge of the search range gets no T_{peak} error and is left out of the fit.



WINDOW (MS)	E_N (S ⁻¹)	T_PEAK (K)	ΔC/C [∞] AT PEAK	ARRHENIUS FIT
0.06 / 0.3	6,706.0	302.38 ± 4.64	-9.77e-3	used
0.12 / 0.6	3,353.0	310.19 ± 2.67	1.04e-2	used
0.24 / 1.2	1,676.5	298.15	-6.43e-3	excluded (edge peak)
0.6 / 3	670.6	309.31 ± 1.52	-9.74e-3	used
1.2 / 5.88	339.6	302.71 ± 9.13	-8.67e-3	used

Arrhenius result in the two tabs

Same transients, same windows, same search range, smoothing spline. Detailed Analysis' figure is its Standard block.

TAB	ET	Σ (CM ²)	R ²	USED + EXCL.	NT (CM ⁻³)
Quick	-1.036 ± 2.261 eV	8.8e-41 ± 7.5e-39	-0.142	4 + 1	5.564e+12
Detailed (Standard)	-1.036 ± 2.261 eV	8.8e-41 ± 7.5e-39	-0.142	4 + 1	5.564e+12

07 • FOUND DURING AND AFTER THE RUN

What this run turned up, and what was fixed

Triggered acquisitions put a time axis in the tick-stamp channels

The evaluation, the writer and the first 58 tests assumed `tickStampImps` holds 60 MHz clock ticks. The triggered `pull_data()` path that runs use stores the DAQ grid's time axis there instead: float seconds, 0 to 4.89 s. The writer cast those channels to `uint64`, which truncates every value to 0 through 4. `cleanup_data()` uses that channel to drop samples and rebuild `timeStampImps`, so analysis of a real HDF5 run would have diverged from the JSON run. Reproduced first on the 23 Sep JSON run: both tick-stamp channels collapsed to 2 distinct values.

Fixed before the run

`zurichInstruments_Control.writeDataH5()` stores tick stamps as `uint64` only when they arrive as integers, and as lossless `float64` otherwise. Regression tests use the triggered data shape (writer round trip, JSON-equivalent cleanup, converter).

Extract & Average found no pulses at the GUI's default pulse levels

`liveDataTab._compute_legacy_transients()` located fill pulses with a fixed test, `AuxInput1 > -2.5 V`, which suits 0 / -5 V pulses. The GUI-default Aux settings give -0.5 / -1.5 V, which never cross -2.5 V, so Qualitative Extract & Average and Detailed Analysis silently skipped every file, JSON and HDF5 alike, and the Qualitative plot stayed empty. The Timing Boundaries also kept their 500 ms default although this run's reverse bias is 6 ms, and the plot's "pF" axis showed values in the millions with no unit scaling.

Fixed after the run; this report uses the fixed code

The pulse search now puts its threshold midway between each file's own two excitation levels (-2.5 V for 0 / -5 V pulses, so the 47 files of the 23 Sep run give identical transients to before; -1.0 V here). A file with no pulses, or none followed by a full reverse-bias window, is reported in the log with the reason instead of being skipped silently, and the plot says so. Loading a run folder fills Timing Boundaries from its `runParams.txt` (State Enable Time = reverse bias, State Disable Time = fill). The plot scales its axis to pF, nF, μ F or mF and no longer uses offset notation.

Live Tools emissions had a time axis near 10^{-11} s

`impdData.cleanup_data()` treated the triggered time axis's genuine first sample ($t = 0$ s) as a missing clock tick. It replaced it using a step of `int(1.87e-5) = 0`, which falls back to 1, giving -1 s, and then divided the whole axis by 60 MHz. Emission plots in Live Tools and the time axis Quick Analysis takes from them came out on a $\sim 10^{-11}$ s scale, for JSON and HDF5 runs alike.

Fixed after the run; this report uses the fixed code

`cleanup_data()` recognises a time axis in seconds (float, non-integral) and keeps it in seconds, with its $t = 0$ origin. Integer clock ticks are handled exactly as before. The test suite was 88/88 passing at that point.

Found in the review after the run: run folder, serial link, gaps and temperatures

An empty Data Root Folder gave `init_experiment()` the path `\MMDDYY\HHMMSS\`, a run folder at the root of the current drive; a relative one landed wherever the GUI was started. A controller that stopped answering made `mK2000B.read_temp()` wait forever, which hangs a run. `impdData._gap_remove()` shifted one segment by the wrong offset and did not handle the backward jump an appended record produces. Temperatures parsed from step-file and ZI history names were `int(T) + 273`: decimals dropped (25.5 °C read as 298 K) and 273 instead of 273.15.

Fixed; this report uses the fixed code

`init_experiment()` refuses an empty or relative root folder and builds `Root\MMDDYY\HHMMSS\` with `os.path.join`. The serial port has a 2 s timeout; `read_temp()` retries a missed reply once and then raises `TimeoutError` (lock released), so `go_to_temp()` fails instead of hanging. `_gap_remove()` handles forward gaps, backward jumps and several gaps, on tick and seconds axes. Temperature keys are exact kelvin ($T + 273.15$, decimals kept), rounded to 10^{-6} K.

Found in the review after the run: edge peaks broke both Arrhenius fits

When a window's signal is still rising at the edge of the peak search range, the spline peak finder puts the peak on the last grid point in every bootstrap sample and reports an error of about 10^{-13} K. That point then dominated the error-weighted Arrhenius fit. On the 23 Sep run (47 temperatures) Detailed Analysis with its GUI-default windows gave $E_t = 90.967$ eV, and Quick Analysis with its defaults gave 1.519 eV with an error of 6×10^{10} eV. The two tabs also normalized, skipped and fitted differently, so they disagreed even on good data.

Fixed; section 06 uses the fixed code

A peak on the edge of the search range, or with an error below 10^{-4} K, keeps its T_{peak} but loses its error, and that window is left out of the fit (all windows are fitted unweighted only if fewer than 3 have errors). Quick Analysis now calls Detailed Analysis' own functions for C_{∞} (mean over 40–90 % of the reverse bias), $S(T)$, the peak, the fit and N_t , and both tabs share default windows, search range, γ , N_d and k_B . On the 23 Sep run: Detailed Analysis defaults 0.693 ± 0.013 eV; Quick Analysis defaults 0.708 ± 0.009 eV (two edge windows excluded); with windows suited to the data both tabs give 0.677 ± 0.007 eV, $\sigma = 3.02 \times 10^{-14}$ cm², $N_t = 1.34 \times 10^{13}$ cm⁻³. The test suite is 163/163 passing.

How this was measured

Hardware: Zurich Instruments MFIA DEV32271 and an Instec mK2000B stage. The run used `dltsRun` from `runDlts_Tools` with a stand-in for the GUI parameter widgets, so every hardware call, file name and write went through the application code. Stage temperature was logged every 10 s from a separate thread sharing the controller's serial lock. Live ingest repeated `liveDataTab._ingest_files_async`'s calls on each new file. Afterwards each HDF5 file was re-written as JSON with `writeDataJson()` into the benchmark output folder (never the run folder); both copies went through `impdData`, `_compute_legacy_transients` and `_load_detailed_data`. The DLTS analysis ran Quick Analysis' functions (`dataAnalysisTab._window_peaks`, `_quick_arrhenius`) on the Extract & Average transients and Detailed Analysis' `_compute_detailed_analysis` on its own Load Data of the HDF5 run. Timings are single runs on this PC (Python 3.13.9, h5py 3.12.1, HDF5 1.14.5, numpy 2.1.3) with the OS file cache warm for both formats.